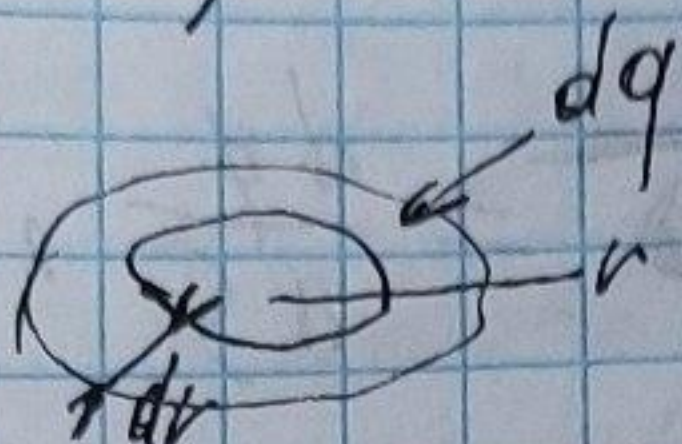


Der normale Radius

$$\int_0^{\infty} dE = \frac{2\lambda z \pi R k}{(R^2 + z^2)^{\frac{3}{2}}}$$

$$q = \lambda 2\pi R$$



$$\begin{cases} dE = \frac{dq k z}{(R^2 + z^2)^{\frac{3}{2}}} ; dq = \frac{Q}{\pi R^2} \cdot 2\pi r \cdot dr = \frac{2Qr dr}{R^2} \\ dq = \sigma dS \end{cases}$$

$$E_z = \frac{2Qr dr k z}{R^2 (R^2 + z^2)^{\frac{3}{2}}}$$

$$\int_0^R dE = \frac{2Qkz}{R^2} \int_0^R \frac{r \cdot dr}{(r^2 + z^2)^{\frac{3}{2}}} = \left| \begin{array}{l} u = r^2 + z^2 \\ du = 2r \cdot dr \\ dr = \frac{du}{2r} \end{array} \right|$$

$$= \frac{2Qkz}{R^2} \int_{z^2}^{R^2 + z^2} u^{-\frac{3}{2}} \cdot \frac{1}{2} du = \frac{Qkz}{R^2} \cdot \left. \frac{u^{-\frac{1}{2}}}{-\frac{1}{2}} \right|_{z^2}^{R^2 + z^2} =$$

$$= \frac{Qkz}{R^2} \cdot (-2) \cdot \left(\frac{1}{\sqrt{R^2 + z^2}} - \frac{1}{z} \right) = -\frac{2Qkz}{R^2} \left(\frac{z}{\sqrt{R^2 + z^2}} - 1 \right)$$

$$\oint_{(S_c)} \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\epsilon_0} ; d\vec{S} = dS \cdot \vec{n} ; \sigma = \frac{dq}{dS}$$

$$d\phi = \vec{E} \cdot d\vec{S} = E dS \cos \alpha ; E S = \lambda \frac{H}{\epsilon_0}$$